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Likelihood Ratio Tests

Until now, in hypothesis testing, we have calculate a statistic t, F or Hotelling's T^2, Today we will learn a more general approach called the 'likelihood Ratio Tests'.

Under H_0 : $H_0 : \theta \in \Theta_0$

Under the full model: $\theta \in \Theta$

where, $\Theta_0 \subseteq \Theta$

The null hypothesis puts restrictions on the possible values of the parameters. The alternative allows more freedom. The likelihood ratio test compares these two situations.

eg: Suppose we have a normal dist^n:

Under the full model: μ can be any value

$\Rightarrow \mu > 0$

But under $H_0 : \mu = 10$

we impose a restriction. μ must be 10.

Which model should fit the data better: the unrestricted model, where μ can take any value or the restricted model, where we force $\mu = 10$? $\longrightarrow$ The unrestricted model will

This is the key idea behind LRT.

Under the full model, maximize: $L(\theta)$ over all possible parameter values. $\sup_{\theta \in \Theta} L(\theta)$

Under the null hypothesis, maximize the likelihood subject to the restriction in H_0.

$$\sup_{\theta \in \Theta_0} L(\theta).$$

Now compare them.

Likelihood Ratio.

$$\Lambda = \frac{\text{Maximum likelihood under } H_0}{\text{Maximum likelihood under the full model}}$$

$$= \frac{\text{restricted}}{\text{Unrestricted}}$$

$$\boxed{\Lambda = \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)}}$$

Because the unrestricted model has more freedom,

$$\sup_{\theta \in \Theta_0} L(\theta) \leq \sup_{\theta \in \Theta} L(\theta)$$

Therefore, $\boxed{0 \leq \Lambda \leq 1}$

Note:

1) If H_0 is true or close to true.
The restriction does not hurt the fit very much. $L_0 \approx L_{full} \Rightarrow \boxed{\Lambda \approx 1}$
So we do not reject H_0.

2) If H_0 is not supported – Forcing the restrictions makes the likelihood much smaller –

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$L_0 \ll L_{full} \Rightarrow$ Therefore $\boxed{\Lambda \approx 0}$

So we have evidance against H_0.

"The smaller the likelihood Ratio, the stronger the evidence against the H_0".

Why we are studying the LRT in Multivariate analysis? Because many multivariate hypotheses impose restrictions on covariance matrix, mean vectors, correlations & ~~so on~~. other multivariate parameters. Hotelling's T^2 was designed mainly for testing mean vectors, but LRT gives us a much more general framework.

$$\left.\begin{array}{l} H_0: \mu = \mu_0 \\ H_0: \Sigma = \Sigma_0 \end{array}\right] \rightarrow \text{Restrictions}$$

or equality of covariance matrices across populations.

Let $X_1, X_2, \ldots X_n$ be a random sample of size n from a p-variate normal dist^n $N_p(\underset{\sim}{\mu}, \Sigma)$

The likelihood function is -

$$L(\underset{\sim}{\mu}, \Sigma) = \frac{1}{(2\pi)^{np/2}|\Sigma|^{n/2}} \exp\left[\frac{-1}{2}\sum_{j=1}^{n}(X_j - \underset{\sim}{\mu})'\Sigma^{-1}(X_j - \underset{\sim}{\mu})\right]$$

Under the full parameter space $\Theta = \{(\mu, \Sigma)\}$ the MLE's are -

$$\hat{\mu} = \bar{x} = \frac{1}{n}\sum_{j=1}^{n} x_j \quad \text{and} \quad \hat{\Sigma} = \frac{1}{n}\sum_{j=1}^{n}(x_j - \bar{x})(x_j - \bar{x})' \qquad \text{---(1)}$$

Simplifying the Exponent of MLE,

Using the trace trick

Let A be a $k \times k$ symmetric matrix & x be a $(k \times 1)$ vector then -

$$x'Ax = tr(x'Ax) = tr(Axx')$$

Using this result in the exponent term of MND, we get

$$\sum_{j=1}^{n}(x_j - \bar{x})'\hat{\Sigma}^{-1}(x_j - \bar{x}) = \sum_{j=1}^{n} tr\left(\hat{\Sigma}^{-1}(x_j - \bar{x})(x_j - \bar{x})'\right)$$

$$= tr\left[\Sigma^{-1}\sum_{j=1}^{n}(x_j - \bar{x})(x_j - \bar{x})'\right]$$

Since, $\sum_{j=1}^{n}(x_j - \bar{x})(x_j - \bar{x})' = n\hat{\Sigma}$ [Using eq^n (1)]

$$= tr\left[\hat{\Sigma}^{-1} \cdot n\hat{\Sigma}\right] = tr\left[n \cdot I_p\right] = n \cdot tr(I_p)$$

$$= np$$

Substituting this back into the likelihood function gives the unrestricted max-likelihood-

$$\max_{\underline{\mu}, \Sigma} L(\underline{\mu}, \Sigma) = L(\hat{\mu}, \hat{\Sigma})$$

$$= \frac{1}{(2\pi)^{np/2}|\hat{\Sigma}|^{n/2}} \exp\left[\frac{-np}{2}\right] \qquad \text{---(2)}$$

Now, Restricted maximization Under H_0.

We test the $H_0: \mu = \mu_0$ V/S $H_1: \mu \neq \mu_0$

Under H_0, μ is fixed at μ_0. The likelihood becomes-

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$$L(\underset{\sim}{\mu}_0, \Sigma) = \frac{1}{(2\pi)^{np/2}|\Sigma|^{n/2}} \exp\left[\frac{-1}{2}\sum_j (X_j - \mu_0)' \Sigma^{-1} (X_j - \underset{\sim}{\mu}_0)\right]$$

Under H_0, the MLE for Σ is –

$$\hat{\Sigma}_0 = \frac{1}{n}\sum_j (X_j - \underset{\sim}{\mu}_0)(X_j - \underset{\sim}{\mu}_0)'$$

Simplifying the exponent under H_0:

Applying the exact same trace trick with $\hat{\Sigma}_0$ –

$= np$

The restricted maximum likelihood is –

$$\max_{\Sigma} L(\underset{\sim}{\mu}_0, \Sigma) = L(\underset{\sim}{\mu}_0, \hat{\Sigma}_0) = \frac{1}{(2\pi)^{np/2}|\hat{\Sigma}_0|^{n/2}} \exp\left(\frac{-np}{2}\right)$$

Now, constructing the likelihood Ratio statistic ——— (3)

$$\Lambda = \frac{\max_{\Sigma} L(\underset{\sim}{\mu}_0, \Sigma)}{\max_{\mu, \Sigma} L(\underset{\sim}{\mu}, \Sigma)} = \frac{L(\underset{\sim}{\mu}_0, \hat{\Sigma}_0)}{L(\hat{\underset{\sim}{\mu}}, \hat{\Sigma})}$$

Substituting eq^n (2) & (3):

$$\Lambda = \frac{\frac{1}{(2\pi)^{np/2}|\hat{\Sigma}_0|^{n/2}} \exp\left(\frac{-np}{2}\right)}{\frac{1}{(2\pi)^{np/2}|\hat{\Sigma}|^{n/2}} \exp\left(\frac{-np}{2}\right)}$$

$$\Lambda = \frac{|\hat{\Sigma}|^{n/2}}{|\hat{\Sigma}_0|^{n/2}} = \left[\frac{|\hat{\Sigma}|}{|\hat{\Sigma}_0|}\right]^{n/2}$$

The equivalent statistic $\Lambda^{2/n} = \frac{|\hat{\Sigma}|}{|\hat{\Sigma}_0|}$ is called Wilk's lambda. If the observed value of the likelihood ratio is too small, the $H_0: M = M_0$ is rejected.

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Confidence Regions

Confidence interval in the univariate case –

Suppose $X_1, X_2, \ldots, X_n \sim N(\mu, \sigma^2)$

We want to estimate μ. From the sample we calculate $\bar{X}$. Does one value, $\bar{X}$, tell us exactly where μ is? NO

So, we give a range of plausible values for μ. ⟶ Confidence interval

So far, we have studied estimation and hypothesis testing. We have estimated unknown parameters and we have tested hypotheses about them.

Now we will discuss another important concept called confidence intervals & Confidence regions.

Point estimate of $\mu = \bar{X}$

Instead of giving one estimated value, we can give a range of possible values within which the unknown (true) parameter is likely to lie? This leads to confidence interval.

"A confidence interval gives us a range of plausible value for an unknown par population parameter".

Suppose, $X_1, X_2, \ldots X_n \sim N(\mu, \sigma^2)$

We want to estimate μ.

Our point estimate is $\hat{\mu} = \bar{x}$

But rather than reporting only $\bar{x} = 50$

We may report: $\mu \in [46, 54]$

Eg (2) Suppose we want to estimate the avg. height of the students in a college.

We take a random sample of $n = 10$ students and obtain

$\bar{x} = 170$ cm, $s = 6$ cm (σ is unknown)

We know the sample mean is 170 cm. But can we conclude that the population mean is exactly 170 cm? → NO

The sample mean will change from sample to sample.

So, instead of giving one estimate, we want a range of plausible values for the true population mean μ.

(then introduce)

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

General $100(1-\alpha)\%$ confidence interval

$$P\left[-t_{\alpha/2, n-1} \leq \frac{\bar{x} - \mu}{s/\sqrt{n}} \leq t_{\alpha/2, n-1}\right] = 1 - \alpha$$

CR-2

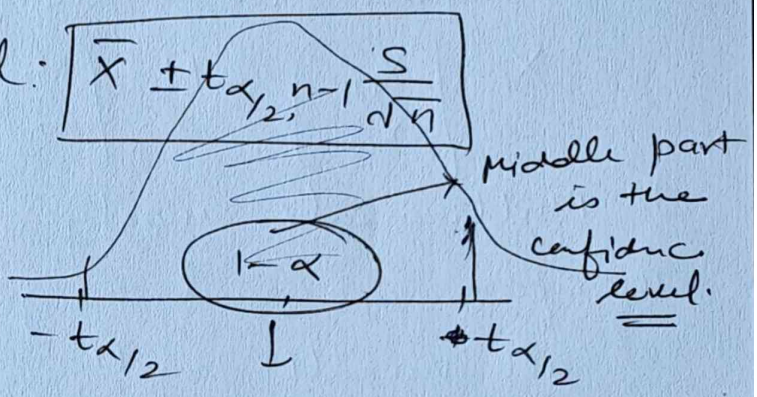
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we obtain, $\bar{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}} \le \mu \le \bar{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$

Or more compactly -

confidence interval: $\boxed{\bar{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}}$

$(1-\alpha)$ represents the prob. in the central region.



95% confidence:

0.05

$\alpha = 0.05$

$100(1-\alpha)\%$ confidence region

$1-\alpha = 1-0.05 = 0.95 \Rightarrow 100(1-\alpha)\% = 95\%$

The small areas outside are the values that our confidence procedure does not cover.

" If we repeatdly take many random samples & construct a 95% confidence interval using the same method, approximately 95 out of every 10 intervals will contain the true population mean.

95% confidence intervals out of 100 contain μ.

& approximately 5 confidence intervals out of 100

for $100(1-\alpha)\%$ → Confidence percentage miss μ.

So far we have learned only one variable.
Suppose we want to estimate the population mean μ

$$\bar{X} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

Here μ is only one number. Therefore all possible value of μ lie on a number line

What happens when we have more than one var?

$$X_i = \begin{bmatrix} X_{i1} \\ X_{i2} \end{bmatrix}, \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

In Multivariate analysis, the var. may be correlated. Therefore, we want one joint region that covers all variables together.

One dim parameter → Confidence Interval

p-dim par. space → Confidence Region

$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$, $t^2 = \frac{n(\bar{X}-\mu)^2}{s^2}$ → This means how far the unknown population mean μ is from the $\bar{X}$, after accounting for variability!

$T^2 = n(\bar{X}-\mu)' S^{-1} (\bar{X}-\mu)$

This is exactly the same idea, but now we are measuring distance between two vectors instead of two no.'s.

In the univariate case, all values of μ that fall within our acceptable t-range form the confidence interval. In exactly the same way, in multivariate analysis, all mean vectors μ that fall within the acceptable Hotelling's T^2 range

the confidence region.

For a one-sample MN population -

$$\frac{(n-p)}{p(n-1)} T^2 \sim F_{p,n-p}$$

$$T^2 \sim \frac{(n-1)p}{n-p} F_{p,n-p}$$

Therefore we do not reject the proposed μ when,

$$T^2 \leq \frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)$$

Hence the set of all such μ is -

$$n(\bar{X}-\mu)' S^{-1} (\bar{X}-\mu) \leq \frac{p(n-1)}{n-p} F_{\alpha,n-p}(\alpha)$$

and this is the $100(1-\alpha)\%$ confidence region for the population mean vector.

$\delta_1 = \mu_{A1} - \mu_{B1}$

$\delta_2 = \mu_{A2} - \mu_{B2}$

Qus:- Age and Risky behaviour are the two important factors that make diffrnce between accident group (AG) & Non-accident group (NAG) of workers. Random samples of 20 individuals from AG and 50 individuals from NAG were collected. The sample mean vector & samp covariance matrix are given. Construct 95% CR for the diff between the two population mean vectors.

Sample 1

$$\bar{X}_A = \begin{bmatrix} 50 \\ 6 \end{bmatrix}, \quad S_A = \begin{bmatrix} 16 & -5 \\ -5 & 4 \end{bmatrix}$$

Sample 2

$$\bar{X}_B = \begin{bmatrix} 40 \\ 8 \end{bmatrix}, \quad S_B = \begin{bmatrix} 25 & -6 \\ -6 & 9 \end{bmatrix}$$

Sol: $n_A = 20$, $n_B = 50$, We want to construct a 95% confidence Region for - $\mu_A - \mu_B$ $\quad \delta = \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix}$

$$S_p = \begin{bmatrix} 22.4853 & -5.7206 \\ -5.7206 & 7.6029 \end{bmatrix}$$

$$T^2 = \frac{n_A n_B}{n_A + n_B} \left[(\bar{X}_A - \bar{X}_B) \right]' S_p^{-1} (\bar{X}_A - \bar{X}_B)' \Big]$$

For a 95% CR, $T^2 \leq \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1 + n_2 - p - 1}$ (0.95)

3.134

$\alpha = 0.05$

$$T^2_{critical} = \frac{2\times 68}{67} \times 3.134 \cong 6.361$$

$$\bar{X}_1 - \bar{X}_2 = \begin{bmatrix} 10 \\ -2 \end{bmatrix} \quad T^2_{obs} \leq$$

$$\{(\bar{X}_1 - \bar{X}_2) - \delta\}' S_p^{-1} \{\bar{X}_1 - \bar{X}_2 - \delta\} \leq c^2$$

where $\delta = \mu_1 - \mu_2$

$$c^2 = T^2 \leq [(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)]' S_p^{-1} [(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)]$$

$$\begin{bmatrix} 10-\delta_1 \\ -2-\delta_2 \end{bmatrix}' \begin{bmatrix} 0.0550 & 0.0414 \\ 0.0414 & 0.1627 \end{bmatrix} \begin{bmatrix} 10-\delta_1 \\ 10-\delta_2 \end{bmatrix} \leq 3.134$$

$$\frac{n_1 n_2}{n_1 + n_2}$$

$$\begin{bmatrix} 10-\delta_1 & -2-\delta_2 \end{bmatrix} \begin{bmatrix} 0.0550 & 0.0414 \\ 0.0414 & 0.1627 \end{bmatrix} \begin{bmatrix} 10-\delta_1 \\ 10-\delta_2 \end{bmatrix}$$

$$\leq 3.134$$

This is the C.R

Solve it